

Sarah Connor

AI Security Engineer · Adversarial ML & Red Teaming

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SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

EXPERIENCE



Senior AI Security Engineer

Cyberdyne Systems

2023 – Present

Austin, TX (US)

Leading red-team engagements against internal LLM-powered products before launch.

- Built an automated prompt-injection fuzzing harness that surfaced 40+ high-severity jailbreaks across 6 production LLM features prior to release.
Custom mutation engine over a 12k-seed corpus, wired into the release-gate CI job; findings triaged weekly with the app-sec team.
- Led adversarial robustness reviews for a fraud-detection model serving 2M inference requests/day, reducing evasion success rate from 18% to under 2%.
Combined gradient-based attack simulation with a red-team-authored adversarial test suite; findings drove a retraining cycle with hardened features.
- Designed and shipped a model-extraction detection pipeline (query-pattern anomaly scoring) now running across all customer-facing inference endpoints.

Python · PyTorch · Burp Suite · Kubernetes · OPA/Gatekeeper



Security Engineer, Applied ML

Skynet Analytics

2021 – 2023

Remote (US)

Owned the security review process for the ML platform team’s model deployment pipeline.

- Authored the org’s first threat model for LLM-integrated services, adopted as the standard review template company-wide.
- Cut mean time to patch critical model-serving CVEs from 21 days to 4 by embedding security gating directly into the MLOps CI/CD pipeline.
- Ran quarterly red-team exercises against internal chatbots, discovering data-exfiltration paths later closed via output filtering and scoped tool permissions.

Terraform · AWS · MLflow · Semgrep · Snyk



Machine Learning Engineer

Tyrell ML

2019 – 2021

Ljubljana, Slovenia (EU)

Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.

- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company’s core payments pipeline.
- Designed the team’s first automated retraining pipeline, cutting model refresh time from weeks to days.

TensorFlow · scikit-learn · Docker

SKILLS

Security	Threat modeling, Burp Suite, Nmap, Metasploit
ML/AI	PyTorch, vLLM, adversarial robustness, prompt-injection defense
Infra	Kubernetes, Terraform, AWS, CI/CD pipeline hardening
Practice	Incident response, vulnerability research, red-team exercise design

> EDUCATION



M.S. Computer Security

Stanford University

2023

Stanford, CA



B.S. Computer Science

University of Ljubljana

2019

Ljubljana, Slovenia (EU)

> PROJECTS



promptfirewall

github.com/sconnor/promptfirewall

2023

> CERTIFICATIONS, AWARDS & PUBLICATIONS



OSCP – Offensive Security Certified Professional

Offensive Security

2019



AI Village CTF – 1st Place

DEF CON

2024



Detecting Prompt Injection at Scale

arxiv.org/abs/2403.09217

2024